KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

DEPARTMENT OF GEOLOGICAL ENGINEERING

PROJECT REPORT

ASSESSMENT OF POTENTIAL HEAVY METAL POLLUTION OF GROUNDWATER IN A MINING COMMUNITY- A CASE STUDY FROM HIA, AMANSIE CENTRAL DISTRICT.

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ABSTRACT

Heavy metal pollution of groundwater presents serious environmental and public health challenges, especially in industrial areas or active mining (illegal mining) communities. This study was conducted to determine the potential heavy metal pollution of groundwater in Hia in Amansie Central District. A total of nineteen groundwater samples were analysed for Physicochemical parameters and heavy metal concentrations. The indices used include the Heavy Metal Pollution index (HPI), Metal Index and Geoaccumulation index(Igeo) and the multivariate statistical analysis such as Cluster Analysis (CA), Principal Component Analysis (PCA). From the results the pH values ranged from 4.04 to 6.56, indicating that the groundwater is slightly acidic, whiles the electrical conductivity (EC) and total dissolved solvents (TDS) were within acceptable limits. This study reveals that the concentrations of Cd, Ni, Zn, Hg, and As do not exceed the WHO limit whereas Fe, Pb, and Mn exceed the WHO limit. Spatial analysis revealed that the highest contamination occurred in the eastern parts of the study area which coincides with intensive small-scale mining (galamsey) The pollution indices indicated consistently that most of the boreholes are moderately to strongly contaminated with BH2 exhibiting very high concentrations. In the correlation assessment, the bivariate correlation proves that heavy metals in the groundwater have independent transport pathways across a defined hydrogeochemical landscape. The cluster analysis also pinpoints significant anthropogenic contamination at (cluster 0; BH 1, BH 3, BH 4) so there should be an element-specific screening.

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CHAPTER ONE: INTRODUCTION

1.1 BACKGROUND

Water is an essential, inorganic, and polar chemical compound (H_20) composed of two hydrogen atoms bonded covalently to one oxygen atom. Water resources are essential to maintaining ecosystem balance and fostering a harmonious development of human society, the environment, and the economy (Sanad et al., 2024). 3% of the total water on earth is freshwater and from this 3%, 95% constitutes groundwater serving as a vital freshwater source valued for its broad accessibility, safety, and cleanliness (Ehsan et al., 2025). Approximately 2.5 billion people globally rely exclusively on groundwater for their daily drinking needs (Sanad et al., 2024). Moreover, groundwater plays an important role in providing water for agricultural and industrial purposes, with about 43% of agricultural water sourced from it (Sanad et al., 2024).

Groundwater is one of the most important freshwater resources for domestic purposes, agriculture, and industrial activities worldwide, particularly in regions facing surface water scarcity and rapid population growth (Kana, 2022; Rezaei et al., 2019). Although groundwater is commonly perceived as safer than surface water due to natural filtration processes within aquifer systems, its quality has deteriorated considerably over recent decades. This decline is largely attributed to intensified anthropogenic activities such as urbanization, industrialization, agriculture, mining, and inadequate waste management practices (Kana, 2022; Venkatramanan et al., 2015). The resulting degradation of groundwater quality poses serious risks to public health, ecosystem integrity, and sustainable socio-economic development (Sanad et al., 2024). Among the various contaminants affecting groundwater systems, heavy metals are of particular concern due to their toxicity, persistence, non-biodegradable nature, and capacity to bioaccumulate in living organisms (Kana, 2022; Venkatramanan et al., 2015).

Heavy metals are naturally occurring elements with high atomic weight and a density at least five times greater than water. Heavy metals frequently reported in groundwater include mercury (Hg), manganese (Mn), nickel (Ni), cadmium (Cd), iron (Fe), zinc (Zn), arsenic (As), and lead (Pb), originating from both natural and anthropogenic sources (Rezaei et al., 2019). Natural inputs primarily result from geological weathering, mineral dissolution, atmospheric precipitation, infiltration of surface runoff, and volcanic activity, whereas anthropogenic contributions are associated with agricultural runoff, industrial effluents, sewage discharge, landfill leachates, mining, and mineral processing activities (Kana, 2022; Venkatramanan et al., 2015).

Assessment of the activities that introduce heavy metals into the environment are of high importance because of its adverse effects on human life, animals and many others since the groundwater is tapped for domestic purposes, and by plants roots for growth. Although some of these heavy metals such as Cr, Pb, Hg, Cd, As, and Co have no useful effects in the body system, long time exposure may cause more acute interruptions in the normal operations of the human organ systems where the metals are deposited (Chaturvedi et al., 2018). Other metals like Zn and Fe for example, are considered micronutrients required for normal growth and functioning of the human body; but at higher concentrations, they become toxic (Dagdag et al., 2023; Kana, 2022).

The Amansie district (Central, West and South) in the Ashanti Region of Ghana are underlain by three geological formations, primarily dominated by the Birimian Supergroup and the Tarkwaian system.

In Ghana, groundwater resources are increasingly threatened by intensive anthropogenic activities, particularly those related to mining activities. . These activities include illegal mining or galamsey, which often lacks adequate environmental management, resulting in the release of potential toxic heavy metals into soils and water bodies, with adverse consequences for ecosystems and human health (Anesthesia et al., 2022; Ekins et al., 2022). Such environmental degradation poses challenges to achieving the United Nations Sustainable Development Goals (SDGs), particularly SDG 3 (Good Health and Well-being), SDGs 6 (Clean Water and Sanitation), SDGs 12 (Responsible Consumption and Production), and SDGs 15 (Life on Land) (Guiraudon, 2005).

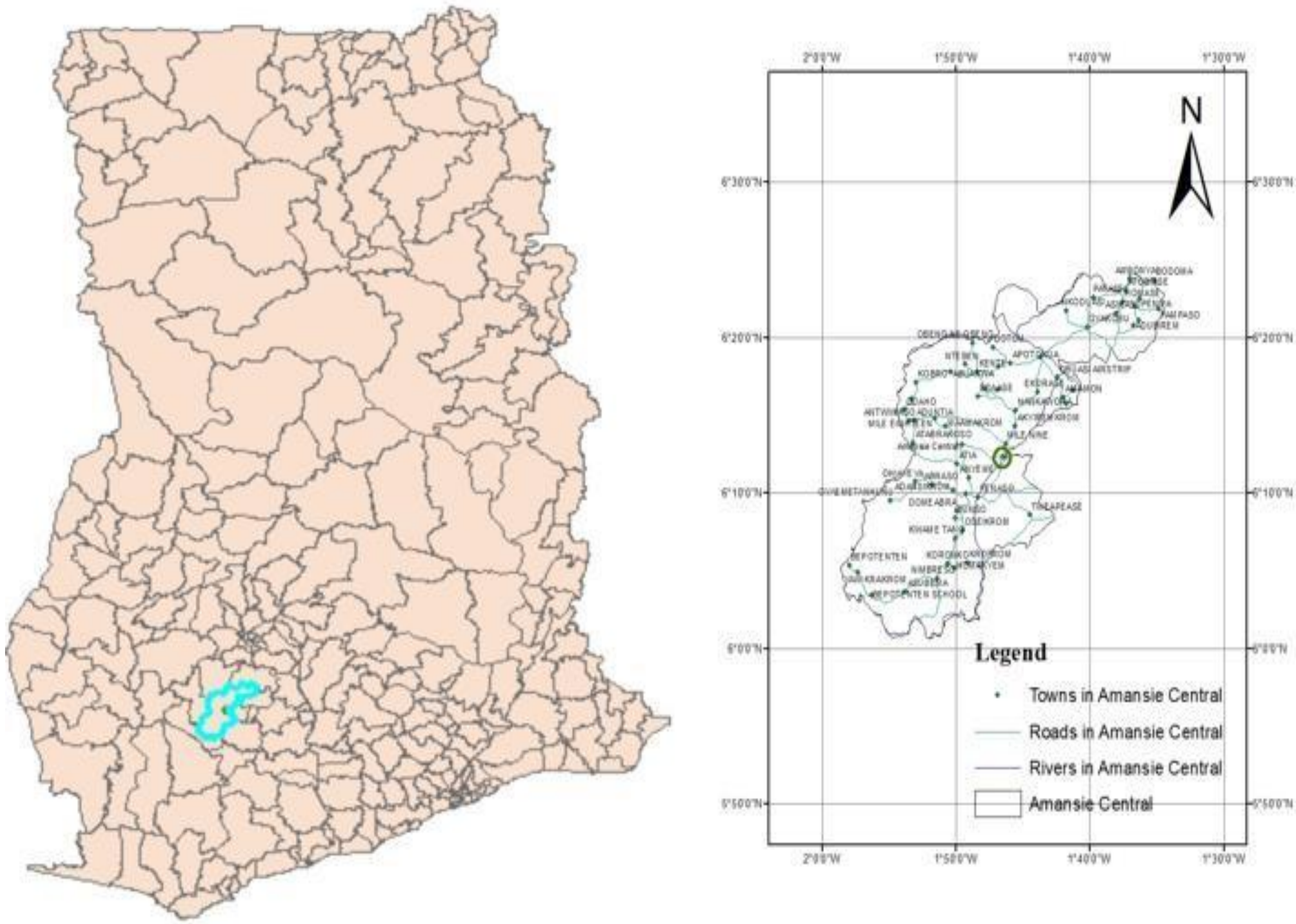


Figure 1: Study area of Hia, Amansie Central.

To evaluate groundwater contamination by heavy metals in the study area, integrated analytical, statistical, and index-based approaches will be used. In this study, pollution indices including the Heavy Metal Pollution Index (HPI), Geoaccumulation Index (Igeo), Metal Index (MI), Degree of Contamination (Cd), and Pollution Index (PI) are employed to assess overall groundwater quality.

The Heavy Metal Pollution Index (HPI) will be used to provide a comprehensive assessment of the overall influence of multiple heavy metals on groundwater quality for drinking purposes. The Geoaccumulation Index (Igeo) will be used to determine the extent of heavy metal accumulation and distinguish between natural geological contributions and anthropogenic sources of contamination. The Metal Index (MI) will be applied to evaluate the cumulative effect of heavy metals and determine the suitability of groundwater for human consumption. The Degree of Contamination (Cd) will be used to quantify the overall level of contamination by integrating the contributions of individual heavy metals, while the Pollution Index (PI) will assess the contamination level of each heavy metal relative to its permissible limit, thereby identifying the metals that pose the greatest environmental and health risks. Collectively, these indices will provide a comprehensive understanding of groundwater contamination, improve the interpretation of the analytical data, and support informed decision-making for groundwater quality management in the study area (Rezaei et al., 2019; Venkatramanan et al., 2015).

In addition, multivariate statistical techniques such as correlation analysis, principal component analysis (PCA), and cluster analysis (CA) are used to elucidate relationships among heavy metals, differentiate between geogenic and anthropogenic sources, and identify controlling hydrogeochemical processes. Geographic Information System (GIS)-based spatial analysis is further applied to enhance the interpretation of spatial contamination patterns and potential pollution hotspots across the Amansie District (Sanad et al., 2024; Sbai et al., 2024).

In addition, multivariate statistical techniques such as correlation analysis, principal component analysis (PCA), and cluster analysis (CA) are used to elucidate relationships among heavy metals, differentiate between geogenic and anthropogenic sources, and identify controlling hydrogeochemical processes. Geographic Information System (GIS)-based spatial analysis is further applied to enhance the interpretation of spatial contamination patterns and potential pollution hotspots across the Amansie District (Sanad et al., 2024; Sbai et al., 2024).

1.2 PROBLEM STATEMENT

Hia, in the Amansie Central District, is a galamsey dominated area. The intensive illegal gold mining poses a significant health risk to groundwater quality through the release of heavy metals into groundwater sources. In addition to mining, agricultural activities also contribute to heavy metal pollution. Heavy metals such as Cd, Fe, Mn, Ni, As, Pb, Hg, and Zn among others are released into groundwater sources due to mining activities as well as agricultural activities. Elevated concentrations of these heavy metals in groundwater endangers both humans and soil organisms. Due to the high dependency on groundwater resources, it is important to implement measures to monitor its quality in such mining or industrial areas. However, such monitoring is largely absent in Hia despite the intensive mining activities (Soceanu et al., 2021). Given the potential risk associated with prolonged exposure to excessive concentrations of heavy metals, there is a need to assess the extent of heavy metal contamination of groundwater in Hia.

1.3 JUSTIFICATION

Groundwater is a vital water source for many communities in the Amansie District of Ghana. Mining (galamsey), industrialization, and agricultural activities are potential sources of heavy metals that can contaminate groundwater. Understanding the extent and sources of these heavy metals is essential for developing effective policies and management strategies to continually monitor and reduce groundwater contamination. This assessment will help protect public health by identifying potential health risks associated with long-term exposure to toxic heavy metals through drinking water. It also supports the achievement of the United Nations Sustainable Development Goals (SDGs), particularly SDG 3 (Good Health and Well-being) by reducing exposure to hazardous substances, SDG 6 (Clean Water and Sanitation) by promoting safe and sustainable groundwater resources, SDG 12 (Responsible Consumption and Production) by encouraging responsible environmental management, and SDG 15 (Life on Land) by contributing to the protection of terrestrial ecosystems from pollution. Overall, the findings will provide valuable information for policymakers, environmental regulators, and local communities to improve groundwater quality management and promote sustainable development in the Amansie District (Ekins et al., 2022) .

1.4 AIM AND SPECIFIC OBJECTIVES

The objective of this project is to evaluate the potential for heavy metals (Mn, Hg, Ni, Cd, Fe, Zn, As, Pb) pollution of groundwater in Hia.

1.5 SPECIFIC OBJECTIVES

The specific objectives are to

(i) determine the levels and spatial distribution of heavy metals in Groundwater,

(ii) determine the degree of heavy metal pollution using various pollution indices and

(iii) Identify the potential sources of heavy metals through multivariate statistical analysis.

CHAPTER TWO: LITERATURE REVIEW

2.1 DESCRIPTION OF THE STUDY AREA

2.1.1 Geographical location and size

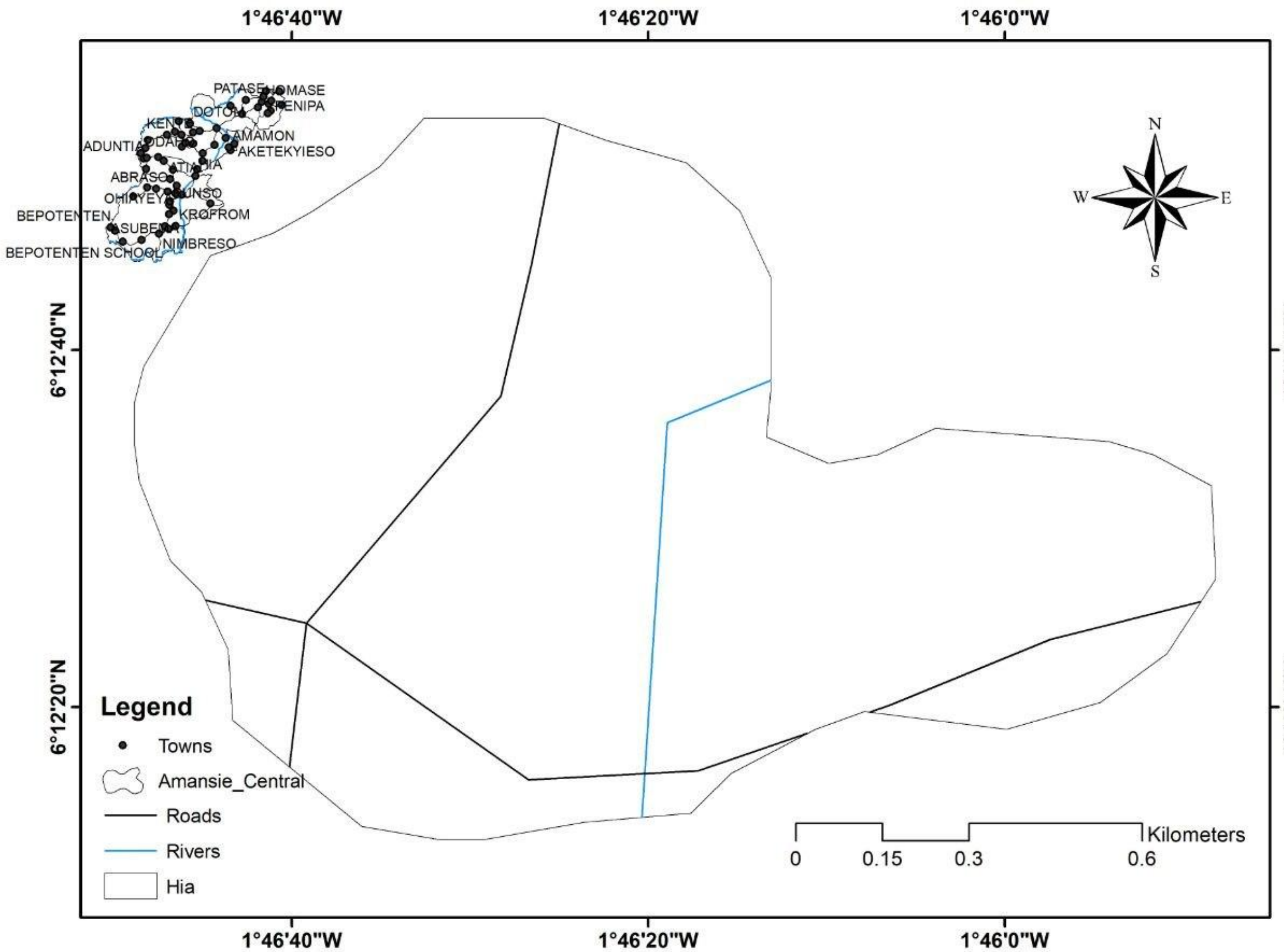


Figure 2: Map of geology of the study area.

The study area, Hia, is located within the Amansie Central District in the Ashanti Region. The district lies approximately between 6° 00’N and 6° 30’N and 1° 15’W and 2° 00’W (Statistical Service, 2022; Tay et al., 2018). It covers a land area of about 710 km^2 with a current population density of 138.7 persons per km2 and has about 206 settlements with Jacobu as the administrative capital (Statistical Service, 2022). Amansie Central District shares common boundaries with Amansie East to the northeast, Amansie West to the west, Obuasi District Assembly to the southeast, Adansi North to the east, Adansi South to the south, and Upper Denkyira in the Central Region to the south (Statistical Service, 2022). The population of the district was recorded as 90,741 which represents the inter-censal growth rate of 1.02% between the years 2000 and 2010. This constitutes 45,466 females and 45,275 males representing 50.2% and 49.8% of the population. The Amansie District is dominated by an agricultural sector which employs about 78% of the labour force in the district (Statistical Service, 2022). However, due to relatively small farm sizes and low yields, poverty among farmers is very high hence they cultivate only few crops and cash crops. In addition to agriculture, artisanal and small-scale mining activities are widespread due to the presence of gold-bearing formations within the district. Water supply in the district is mainly derived from groundwater sources, including small town water systems, boreholes, hand-dug wells and rivers/ streams which serve as the main source of water for both industrial and domestic purposes in the district. Among these sources, about 88% of people rely on boreholes and 9.5% rely on hand dug wells (Statistical Service, 2022). This high dependence on groundwater highlights the importance of assessing its quality, especially in areas affected by mining activities.

2.1.2 Geology

The District is predominantly underlain by the Birimian Supergroup which consists of metasedimentary and metavolcanic units (Tay et al., 2021, 2019). Even though site-specific geological data for Hia is limited, the geology of the area can be inferred from that of the Amansie Central District and surrounding regions as illustrated in (figure. 3). The Tarkwaian System also occurs locally, consisting mainly of sedimentary rocks such as quartzites and conglomerates (Boansi et al., 2021; Tay et al., 2021). The Birimian and the Tarkwaian are the two gold- bearing Lower Proterozoic stratigraphic units in West Africa.

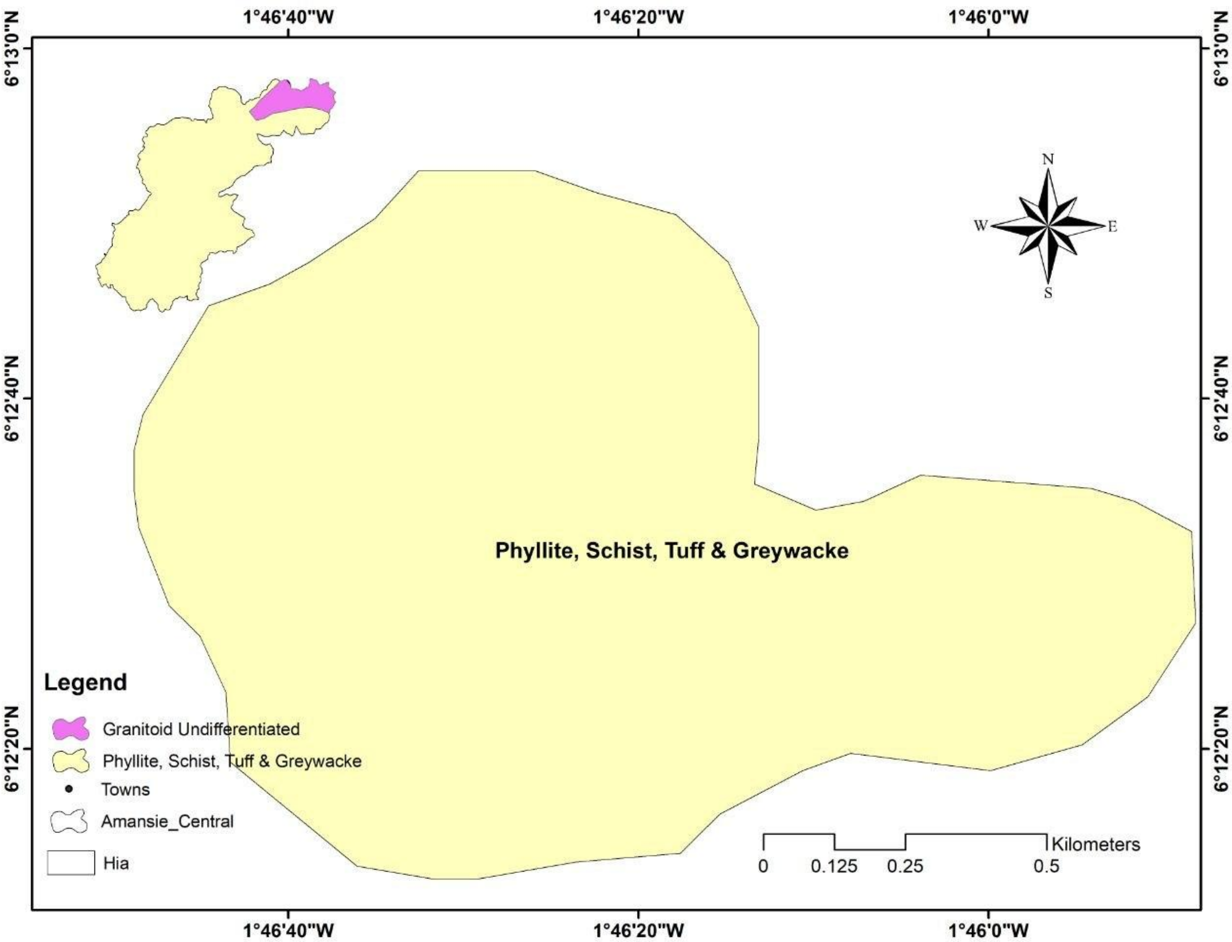


Figure 3:Map of geology of the study area.

The Birimian consists of two distinct lithologic associations that are generally referred to as the Lower and Upper Birimian (Tay et al., 2021). Lower Birimian rocks comprise an assemblage of fine-grained rocks with a large volcanoclastic component. Typical lithologies include tuffaceous shale, siltstone, greywacke and some chemical (Mn-rich) sediments.

Upper Birimian rocks comprise mostly basalts with some interflow sediments. These rocks are predominantly volcanic and pyroclastic (Dzlgbodl-adjlmah, 1993; Leube et al., 1990). The rocks consist of a bedded group of tuffs, sediments and mafic lavas (greenstones), jointly with minor bands of phyllite that include a zone of manganiferous phyllites containing manganese ore. Batholithic masses of granite and gneiss intrude the sequence. The mostly argillaceous sediments in this series have been metamorphosed to schist, slate and phyllite jointly with some interbedded greywacke. A major unconformity separates the Birimian Supergroup from the Tarkwaian system.

The term Tarkwaian refers to clastic sedimentary rocks representing erosion products of the Birimian provenance. The Tarkwaian consists of coarse clastic sedimentary rocks that include conglomerates, arkoses, sandstones and minor amounts of shale. Pebble clasts in the conglomerate include volcanic and sedimentary clasts derived from Birimian rocks and granitoid. The Tarkwaian is usually regarded as the detritus of Birimian rocks that were uplifted and eroded following the Eburnean tectono-thermal event (Dzlgbodl-adjlmah, 1993; Leube et al., 1990).

The geological setting of the area is closely related to gold mineralization. The styles of gold mineralization differ. Birimian deposits are epigenetic gold quartz vein deposit-types, associated with disseminated sulfides in fractured and sheared zones. Tarkwaian gold, on the other hand, is in paleoplacer Banket conglomerates of the Witwatersrand-type (Dzlgbodl-adjlmah, 1993). Tarkwaian deposits are generally interpreted to be derived from the Birimian. This has led to widespread artisanal and small-scale mining activities in areas underlain by these formations. Such mining activities can result in the release of heavy metals into the environment, thereby influencing groundwater quality, particularly in crystalline basement terrains where groundwater occurs within weathered and fractured zones.2.1.3 Hydrogeology and Drainage

Hia is predominantly underlain by the Birimian supergroup which consists of metavolcanic and metasedimentary aquifers as well as the Tarkwaian whereas the Crystalline Basement Granitoid complexes are made up of granitoid aquifer (Boansi et al., 2021). Both aquifers are dominated by low primary porosities and permeabilities. Hydrogeologically, both the Birimian and granitoid aquifers are characterized by low porosity and permeability; secondary porosities are also observed which arises from weathering and the activities associated with the eburnean orogeny (Boansi et al., 2021).

This has led to an increase in the pores and fractures within the aquifers. Groundwater occurrence in the region is controlled by factors such as topography, lithology, and geologic structures. Groundwater occurs in the saprolite, saprock, and fractured bedrock in the Crystalline basement complex. The occurrence of groundwater in the Birimian province is more defined in the lower part of saprolite and the upper part of the saprock. These two regolith profiles are hydraulically connected with their storage and permeability (Boansi et al., 2021). The Birimian exhibits a higher degree of weathering than the granitoids due to the lower jointing and fracturing contained in the granitoids. This has consequently resulted in lower groundwater yields and shallower ground water tables in granitic terrains.

The presence of quartz veins or pegmatites in Birimian Province has been documented to possess higher quantities of groundwater. On the basis of long-term borehole drilling records collated from 1991-2011, the Birimian has an average well yield of 30–60 l/min and a higher drilling success rate (approximately 91%) than the granitoids aquifer with an average borehole yield of less than 30 l/min (Boansi et al., 2021; Osei-Nuamah, I., 2024).The hydrogeological characteristics influence the drainage and recharge processes within area. Hia lies in the Pra River Basin and is drained by a network of local streams that form tributaries of major rivers such the Offin River, which ultimately feeds into the Pra River.

2.1.4 Vegetation

Hia lies in Ghana's (Moist Semi-Deciduous Forest Zone) forest ecological zone. This zone is critical to the country's biodiversity and ecosystem services (Donkor et al., 2023). Moreover, the landscape surrounding Hia has been profoundly altered by both large-scale mining, artisanal and small-scale mining (galamsey).

2.1.5 The regional vegetation context

The Ashanti region's natural vegetation is characterized by tall, multi-layered forest structures. In undisturbed areas, the flora includes high value timber and hardwood tree species such as Odum (Milicia excelsa), Wawa (Triplochiton scleroxylon), Mahogany, Edinam, and Sapele, which are common across the district (Asamoah, 2025). These forests support farming activities, especially cocoa, oil palm, and food crops, making the area largely agrarian and also used as medicines (Asamoah, 2025).

Mining activities in Amansie districts has led to a 304% escalation in land conversion and deforestation, with 30% and 86% of their original forest cover (Üniversitesi et al., 2026). As mining removes the top soil and introduces heavy metal like Mercury (Hg), Arsenic (As), Lead (Pb), which prevents the natural plants regrowth (Glazebrook et al., 2020).

2.1.6 Climate

Hia is characterized by a wet semi-equatorial climate (Tay et al., 2019). This climatic zone is defined by a double rainfall maximum regime, which is critical for the region's socio-economic structure. The major rainy season typically occurs between March and July, followed by a minor rainy season from September to November (Tay et al., 2019).

Quantitative data indicates that the district receives a mean annual rainfall ranging from 855 mm to 1,500 mm (Tay et al., 2019). Other regional profiles suggest total annual rainfall in similar adjacent areas may fall within the 800 to 900 mm range during drier cycles, highlighting internal variability (Ghana Districts, 2025). The thermal regime is remarkably stable, with mean monthly temperatures averaging approximately 27°C (Tay et al., 2019).

2.1.7 Vegetation and Environmental Interaction

The climatic conditions directly support a moist semi-deciduous forest vegetation type, which facilitates the cultivation of major cash crops such as cocoa and citrus, as well as food crops like plantain and vegetables (Tay et al., 2019). However, the natural environment and its role in local climate regulation are under threat. Illegal mining (locally known as galamsey) have significantly degraded the natural forest cover, including vital reserves such as the Odaho, Apanprama, and Aboaboso Forest Reserves (Tay et al., 2019).

2.2 GROUNDWATER QUALITY AND CHEMISTRY

Groundwater quality comprises the physical, chemical, and biological qualities of groundwater. The physical qualities of groundwater include parameters such as temperature, turbidity, colour, taste, and odor whereas groundwater is mostly colourless and odourless hence its biological and chemical properties are of more importance (Harter, 2003).

The usefulness of groundwater for various purposes is determined by its chemical, physical and bacterial characteristics (Ojo et al., 2012). Chemical analysis of groundwater includes the determination of the concentration of inorganic constituents as well as the measurement of pH and specific electrical conductance (EC) (Ojo et al., 2012).

Groundwater contains ions which slowly dissolve from soil particles, sediments, and rocks as the water travels along mineral surfaces in the pores or fractures of the unsaturated aquifer (Harter, 2003). Naturally, groundwater is highly pure and frequently demands little or no treatment be safe (MacDonald & Calow, 2009). However, anthropogenic activities can change the natural composition of groundwater by injecting chemicals and waste into the soil which can significantly alter the natural composition of groundwater. (A. Edet, 2012; Harter, 2003).

2.2.1 Groundwater Physico-Chemical Parameters

Groundwater quality for domestic and agricultural applications is primarily determined by chemical parameters such as pH, Total Dissolved Solids (TDS), Electrical Conductivity (EC), salinity, turbidity, temperature, and aesthetic properties.

pH: Controls the chemical behavior, solubility, and bioavailability of chemical species in groundwater. A pH of 7 represents neutrality, whereas lower indicates acidity and higher values denote alkalinity. The fluctuations in groundwater pH are driven by seasonal hydrological changes, mineral dissolution, and anthropogenic inputs such as agricultural fertilizers and pesticides (Tay et al., 2021).

Total Dissolved Solids (TDS) & Electrical Conductivity (EC): TDS represents the total concentration of dissolved inorganic minerals.

Cations: Ca^{2+}, Mg^{2+}, Na^{+}, K^{+}, A

Anions: HCO_{3}^{-}, CL^{-}, SO_{4}^{2-}, NO_{3}^{-}.

Because dissolved ions directly facilitate electrical transport, EC serves as a direct proxy for ionic strength and total dissolved concentration (Kulshrestha, 2019). Drinking water guidelines generally recommend a maximum TDS threshold of 500 mg/L (EC > 2.25 - 4mS/cm) for aesthetic acceptability, whereas TDS values exceeding 1,500 to2,600mg/L (Harter, 2003).

Salinization: Groundwater salinization impairs soil productivity, destroys aquatic ecosystems, and enhances the mobilization of toxic constituents, including fluoride, arsenic, and selenium (Kulshrestha, 2019).Shallow aquifers in arid and hyper-arid zones are particularly susceptible to salinization driven by intense evapotranspiration, shallow water tables, and unsaturated zone hydraulic properties (Medjani et al., 2021).

Turbidity: Expressed in Nephelometric Turbidity Units (NTU), turbidity measures light scattering caused by suspended sediments, organic matter, and colloidal material (Marfo et al., 2024)High turbidity promotes bacterial proliferation, reduces chlorine disinfection efficiency, and restricts aquatic photosynthesis by attenuation of light penetration (Soceanu et al., 2021)). In fractured or karstic terrains, intense rainfall events accelerate sediment transport directly into groundwater networks (Popoola et al., 2019)•

Temperature, Taste, Odor, and Color: Water temperature influences chemical reaction rates, microbial kinetics, dissolved gas solubility, and volatility-driven odor (Owusu et al., 2024) Water discoloration typically stems from humic substances derived from decomposing surface vegetation, quantified against a standard platinum-cobalt reference solution (1Pt-Co unit = 1 mg/L chloroplatinate ion) (Ojo et al., 2012).Although aesthetic factors such as taste, odor, and color do not always correlate directly with toxicity, they strongly influence public perception and consumer trust in water safety (Riaz et al., 2017)

2.2.2 Quality Control

Quality control constitutes an essential pillar of any methodologically sound research endeavor, as the reliability of a study's conclusions is inherently contingent upon the precision and integrity of its underlying data. Quality assurance encompasses the proactive measures implemented prior to data collection, designed to ensure that established procedures conform to predefined standards from the beginning, whereas quality control involves the evaluative checks conducted during and immediately following data collection to verify that those standards have been effectively upheld (Taşan et al., 2022)Together, quality assurance and quality control (QA/QC) form indispensable components of contemporary analytical geochemistry.

A well-structured QA/QC framework serves to identify sources of analytical error while simultaneously providing a basis for establishing confidence in, and recognizing the limitations of, analytical datasets. Such a program encompasses the continuous monitoring of precision, accuracy, and potential contamination throughout the entire process from sample collection to laboratory analysis. Precision is assessed through the systematic incorporation of sample, pulp, and analytical duplicates alongside certified reference materials, with the resultant data evaluated using scatterplots, statistical tests such as percentage relative standard deviation, Thompson-Howarth plots, and the average coefficient of variation (CV_{avg} (%)). Accuracy is determined through the submission of reference materials and assessed using statistical approaches including percentage relative difference, tests, and Shewhart control charts. Contamination is examined through the use of blanks, the outcomes of which are similarly tracked via Shewhart control charts (Piercey, 2014).

HEAVY METALS

Heavy metal pollution of groundwater is a serious problem for human health, as it causes toxicity and carcinogenic effects (Soceanu et al., 2021)Human health, industrialization, as well as the practices of modern agriculture, require increased attention and protection of groundwater quality but also a rational exploitation within the limits of exploitation reserves (Soceanu et al., 2021). For example, the presence of lead in water beyond the permissible level could result in hypertension, interference with Vitamin D and calcium metabolism, brain development hindrance in foetus and young children, damage to tissues and organs in human and many more (Popoola et al., 2019). In order to assess the groundwater quality, identify sources of contamination, and plan preventive measures to reduce the public health risks like, authorities will need to perform routine monitoring of groundwater (Soceanu et al., 2021).

2.2.3 Heavy Metal Pollution in Groundwater

Heavy metal pollution of groundwater is a major environmental and public health concern, particularly in mining regions such as Hia in the Amansie District in the Ashanti Region of Ghana. Heavy metals are persistent, non-biodegradable, and can accumulate in water over a period of time, posing long-term risks to health (Hossain et al., 2025).

Studies carried out within the Amansie district show that groundwater contains significant concentrations of heavy metals such as iron (Fe), manganese (Mn), arsenic (As), zinc (Zn), copper (Cu), lead (Pb), cadmium (Cd), and mercury (Hg) (Tay et al., 2019). The relative abundance of these metals has been reported in the order Fe > Mn > As > Zn > Cu > Pb > Cd > Hg, indicating dominance of iron and manganese in the groundwater system (C. K. Tay et al., 2019a).These elevated concentrations are attributed to both natural geological processes (such as weathering of mineral-rich Birimian rocks) and anthropogenic activities, particularly artisanal and small-scale mining (galamsey).

Similarly, studies in other Ghanaian mining areas such as Tarkwa have confirmed that groundwater contamination is strongly linked to mining activities, which introduce toxic elements like arsenic and lead into aquifers (Gwira et al.,2023). These metals are of particular concern due to their toxicity and potential to cause severe health effects, including organ damage and increased cancer risk.

In addition, research conducted within Amansie West indicates that mining activities significantly contribute to heavy metal contamination of environmental materials, including soils and groundwater systems (Baah et al., 2023)

Looking at all these studies in nearby locations in the Region or country, it is indicated that heavy metal pollution in groundwater within the Amansie Central District is possibly influenced by both geogenic and anthropogenic factors, with mining activities playing a major role. Hence, there is a need for continuous monitoring and effective management strategies to safeguard water quality.

2.3 POLLUTION INDICES IN GROUNDWATER ASSESSMENT

To effectively evaluate groundwater pollution, researchers have developed various pollution indices that simplify complex hydro-chemical data into easily interpretable values. Among the most commonly used are the Water Quality Index (WQI), Heavy Metal Pollution Index (HPI), and Hazard Quotient (HQ) and Hazard Index (HI) for health risk assessment.

In the Amansie and Adansi districts, pollution evaluation indices such as HQ and HI have been used to assess the potential health risks associated with heavy metal pollution. These indices are based on standard methodologies, such as those developed by the United States Environmental Protection Agency (USEPA) and are used to determine whether groundwater consumption poses significant health risks (C. K. Tay et al., 2019b). The application of these indices allows researchers to quantify both carcinogenic and non-carcinogenic risks associated with contaminated water.

Studies from other mining regions in Ghana, such as Tarkwa, have also applied similar indices to assess groundwater quality. Results show that although some heavy metal concentrations may fall within allowable limits, long-term exposure can still pose health risks, particularly due to bioaccumulation and long-term intake (Seidu & Ewusi, 2020). These findings highlight the importance of using pollution indices as tools for comprehensive water quality assessment.

Pollution indices are particularly useful because they combine multiple parameters into a single value, making it easier to classify water quality into categories such as safe, moderately polluted, or highly polluted. However, some studies suggest that these indices may simplify complex hydrochemical processes too much and should therefore be used alongside other analytical techniques.

2.4 MULTIVARIATE STATISTICAL METHODS IN GROUNDWATER STUDIES

Multivariate statistical techniques have become essential tools in groundwater quality assessment due to their effectiveness to analyse large and complex datasets. Commonly used methods include Principal Component Analysis (PCA) and Hierarchical Cluster Analysis (HCA).

In Ghana, these techniques have been widely applied to identify the sources and processes controlling groundwater chemistry. For instance, a study on groundwater systems in northern Ghana employed PCA and cluster analysis to determine that silicate weathering, agrochemical inputs, and domestic waste were the main factors influencing groundwater quality (Loh et al., 2020). These findings are consistent with studies conducted in the Amansie District, where groundwater chemistry is primarily controlled by silicate weathering and ion-exchange processes (C. Tay et al., 2021).

Principal Component Analysis is particularly useful in reducing data dimensionality and identifying the dominant factors influencing water quality. It helps distinguish between natural processes and anthropogenic influences such as mining. Cluster analysis, on the other hand, groups water samples based on similarities in chemical composition, allowing for the identification of spatial patterns and pollution zones.

The integration of multivariate statistical methods with hydrochemical analysis enhances the interpretation of groundwater data and improves the identification of contamination sources. These techniques are especially important in complex environments like Hia, where multiple factors possibly influence groundwater quality.

Despite their advantages, multivariate methods require careful interpretation and a solid understanding of the study area. When used alongside pollution indices and conventional hydrochemical techniques, they provide a comprehensive framework for groundwater quality assessment.

CHAPTER 3: METHODOLOGY

3.1 DESCRIPTION OF STUDY AREA

The research was done at Hia in the Amansie Central District of Ashanti region. It is situated about 11km W of Obuasi and about 56 kilometers from Kumasi, the regional capital.

The district is underlain by three geological formations. These are the Birimian, Tarkwaian and Granite rocks which are rich in mineral deposits. There are five main classifications of soil in the area. They include Bekwai-Oda, Mim-Oda, Asikuma-Atewa Ansum/Oda, Kumasi-Asuansi/Nsuta Offin and Birim-Awaham/Chichiwere compound Associations, as well as sand and gravel deposits (Ghana Statistical Service)

Gold is abundant in the district and mostly located at Apitisu, Amamom, Anyankyiremu, Adubrim, Fiankoma, Jacobu and Aketechieso which are all in close proximity to Hia (Ghana Statistical Service, 2022).

3.2 SAMPLING COLLECTION AND ANALYSIS

For the purpose of this report, secondary data obtained from Kwame Nkrumah University of Science and Technology (KNUST) was utilized. This data includes 19 borehole samples as shown below in Figure 4 which were collected, taking into consideration how close the inhabitants are close to the illegal mining sites. The data includes the measured concentrations of heavy metals in this study. In-situ physicochemical parameters including pH, Total Dissolved Solids (TDS), Electrical Conductivity (EC)and Salinity was also obtained.

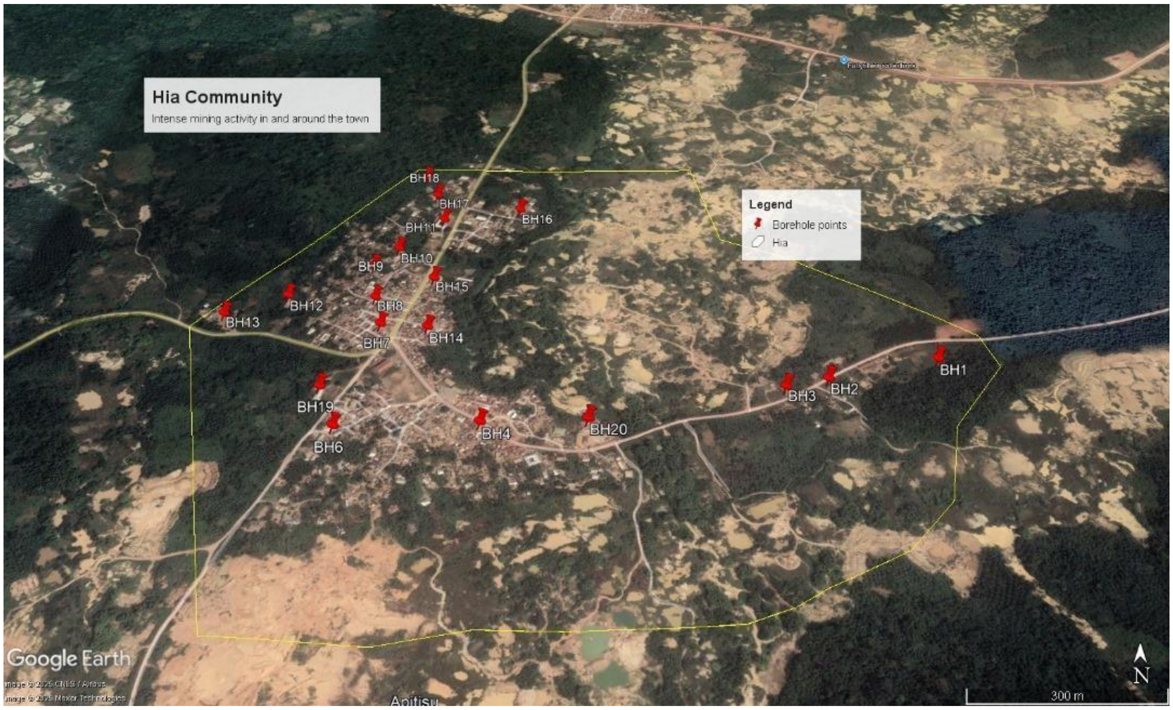


Figure 4: Satellite image of Hia showing the borehole points

3.3 POLLUTION EVALUATION INNDICES

3.3.1 Heavy Metal Pollution Index (HPI)

The Heavy Metal Pollution Index (HPI) is a widely used water quality index developed by Mohan et al. (1996) to evaluate the overall influence of heavy metals on drinking water quality.

The mathematical expression is:

$$HPI = \frac{\sum_{i=1}^{n} WiQi}{\sum_{i=1}^{n} Wi}$$

where:

- HPI = Heavy Metal Pollution Index
- Wi= Unit weight of the *i*th heavy metal
- Qi = Sub-index (quality rating) of the *i*th heavy metal
- n = Number of heavy metals analysed

The quality rating is calculated as:

$$Qi = \frac{|Mi - Ii|}{Si - Ii} x100$$

where:

- Mi= Measured concentration of the *i*th metal (mg/L)
- Ii= Ideal value of the *i*th metal (usually **0 mg/L** for most toxic heavy metals)
- Si= Standard permissible limit of the *i*th metal according to WHO or national drinking-water standards

The unit weight is computed as:

$$Wi = \frac{K}{Si}$$

where:

- K = Constant of proportionality

The constant K is given by:

$$K = \frac{1}{\sum_{i=1}^{n}(\frac{1}{Si})}$$

Thus, metals with lower permissible limits receive higher weights in the HPI calculation. The following Table 2. Show the classification of HPI.

Table 1. Classification of the HPI (Edet & Offiong, 2002).

HPI Values	HPI Classification
<100	Low water pollution
100	Medium water pollution
>100	High water pollution

3.3.2 Metal Index

The Metal Index (MI) is a simple but widely used index for assessing the cumulative effect of heavy metals in water. It was proposed by Tamasi and Cini (2004) and has since been extensively applied in groundwater quality studies.

The mathematical expression is:

$$MI = \sum_{i=1}^{n} \frac{Ci}{MACi}$$

where:

- **MI** = Metal Index
- Ci = Measured concentration of the *i*th metal (mg/L)
- MACi = Maximum Allowable Concentration (or Maximum Admissible Concentration) of the *i*th metal according to WHO or the relevant drinking-water standard (mg/L)
- n = Number of heavy metals considered

The MI is calculated by summing the ratios of the measured concentration of each metal to its corresponding permissible limit as classified in Table 3.

Table 2. Classification of the MI

MI Values	MI Classification
<0.3	Very pure
0.3 < MI < 1	Pure
1 < MI < 2	Slightly affected
2 < MI < 4	Moderately affected
4 < MI < 6	Strongly affected
>6	Seriously affected

3.3.3 Geochemical Index (Igeo)

Geochemical index (Igeo) was originally stated by Muller (1969) in order to determine and define metal contamination in sediments by comparing current concentrations with pre- industrial levels, Igeo is calculated as follows:

$$I_{geo} = \log_2[C_{n/1.5B_n}]$$

where:

Cn is the measured concentration in the sediment for the metal n

Bn is the background value for the metal n

the factor 1.5 is used because of possible variations of the background data due to lithological variations

Table 3. Classification of Geoaccumulation Index (Igeo)

Class	Value	Classification
0	<0	Uncontaminated
1	0-1	Uncontaminated to moderately contaminated
2	1 to 2	Moderately contaminated
3	3 to 4	Moderately to strongly contaminated
4	4 to 5	Strongly contaminated
5	5 to 5	Strongly to extremely contaminated
6	>5	Extremely contaminated

3.4 MULTIVARIATE STATISTICAL ANALYSIS

Microsoft Excel was used for the Principal Component Analysis, Cluster Analysis and Correlation Analysis. These statistical analyses were performed to determine the relationship between the various metals to predict their possible sources.

3.5 SPATIAL DISTRIBUTION ANALYSIS

ArcGIS was used to analyze the spatial distribution of heavy metals in groundwater. The inverse distance weighting (IDW) interpolation method was utilized to generate these maps, illustrating spatial variations throughout the study area. IDW is a basic yet widely applied interpolation technique that enables the estimation of heavy metal concentrations based statistical relationships among known locations (El-Zeiny et al., 2019).

CHAPTER 4: RESULTS AND DISSCUSSION

4.1 PHYSICAL PARAMETERS OF GROUNDWATER

The Physico-chemical parameters that were measured in-situ at the sampling in the study area include pH, Electrical Conductivity, Salinity and Total Dissolved solids (Table 4). In Table 4 below the pH ranged from 4.04 to 6.56 with a mean of 5.07. The recorded pH values indicates that groundwater in the study area is slightly acidic. Total Dissolved solids ranged from 30.9 to 341ppm with an average of 113.46ppm. The TDS values are considered acceptable regarding the palatability of water indicating that there has been no dissolution of salt minerals (WHO 2017), Assessing spatial distribution of heavy metal contamination in groundwater). Electrical Conductivity ranged from 43.4 to 481s/cm with an average of 159.86 which gives an indication of the ionic content likely resulting from industrial effluents (Assessing spatial distribution of heavy metal). Salinity for almost all the water samples was 0ppt with a maximum of 0.1ppt.

Table 4:Physical Parameters measured in-stiu in the field

ID	PH	EC(s/cm)	Salinity(ppt)	TDS (ppm)
BH1	5.73	174	0	123
BH2	5.85	130.2	0	92.8
BH3	5.91	233	0	166
BH4	5.04	255	0.1	181
BH6	4.18	69.1	0	49.1
BH7	5.15	113	0	80.4
BH8	5.13	128.3	0	90.8
BH9	4.38	74.4	0	53
BH10	4.86	98.8	0	70.3
BH12	4.7	271	0.1	192
BH13	5.5	131.4	0	92.8
BH14	5.18	247	0.1	176
BH15	4.57	104.9	0	74.4
BH16	4.42	43.4	0	30.9
BH17	4.28	101.2	0	71.5
BH18	4.04	113.8	0	80.5
BH19	5.74	108	0	76.7
BH20	6.56	481	0.2	341
Mean	**5.07**	**159.86**	**0.03**	**113.46**

The EC values in Figure 5 below exhibit a linear relationship with the TDS values which demonstrates the linear correlation between specific conductance and ionic strength (Kana, 2022). Classification based on heavy metal load and pH (Edet &Offiong, 2002), are also illustrated in figure 6.

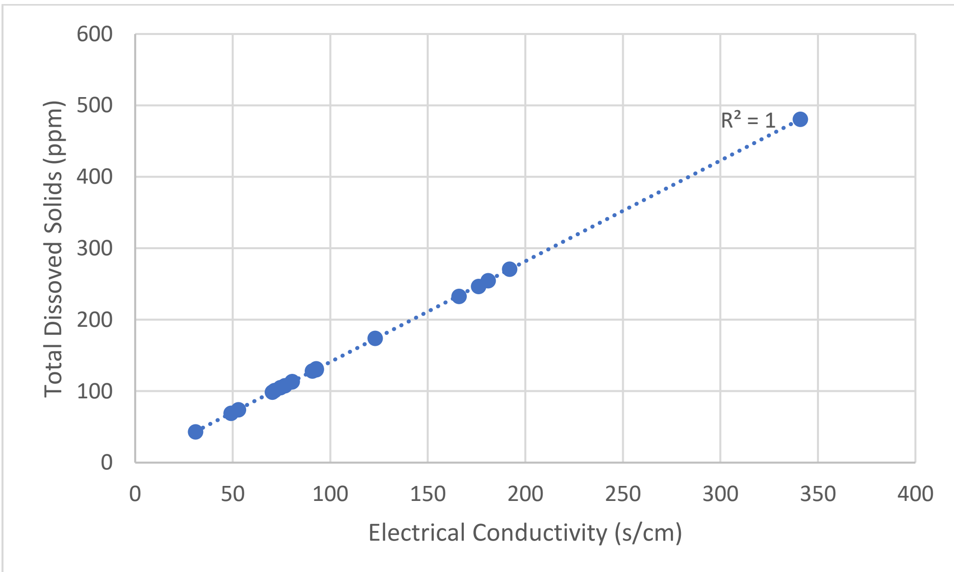


Figure 5.Total Dissolved Solids against EC

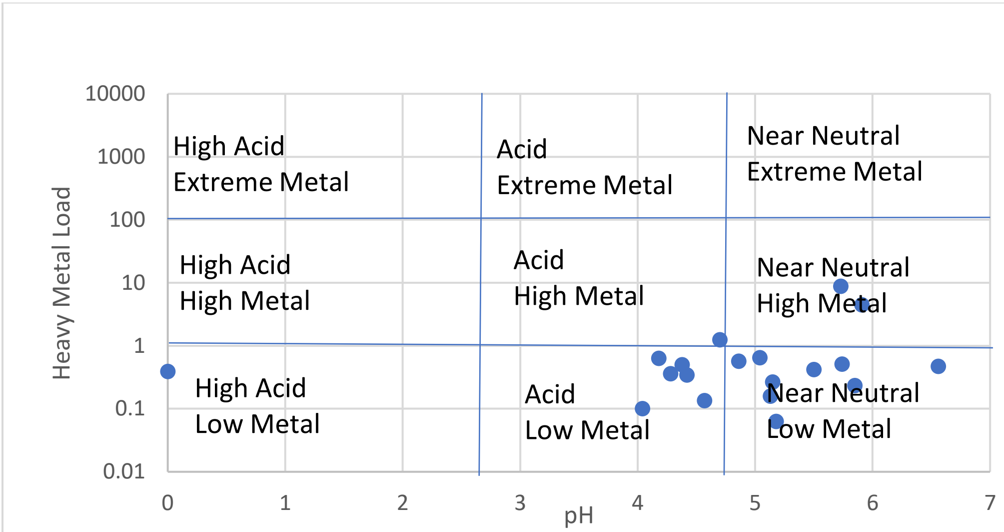


Figure 6: Classification based on heavy metal load and pH (Edet &Offiong, 2002)

4.2 TRACE ELEMENTS IN GROUNDWATER SAMPLES

The concentrations of the various metals determined in the groundwater from the study are shown in figure 6 below. The average concentrations of Cd, Fe, Mn, Ni, Pb, Zn, As, Hg are 0.01µg/L, 933.41µg/L, 81.31µg/L, 8.29µg/L, 15.85µg/L, 25.11µg/L, 2.85µg/L, 5.0µg/L respectively. Cd has a concentration of 0.01µg/L indicating that it is evenly distributed in the study area and it is not in concentrations that exceed WHO limit. The concentrations of Iron (Fe) in eleven of the 19 sampled points exceed the limit of 300µg/L. The presence of Fe in such high concentrations could be as a result of Fe been introduced by natural factors into groundwater even though anthropogenic sources may also introduce Fe into the groundwater. 2 out of the 19 sample points had Mn exceed the allowable limit based on WHO standards. This may be because those sample points are closer to the anthropogenic sources introducing these heavy metals into the groundwater. 11 out of the 19 sample points of Pb concentrations exceeds the WHO limit. None of the samples had Cd, Ni, Zn and As concentrations exceeding the WHO limit.

From Figure 6 above, which shows the classification of groundwater based on its pH and the metal load, which is defined as the sum of the individual metal concentrations at each sample point (Kana, 2022). According to this classification scheme, most of the groundwater quality is predominantly characterized by low metal loads under mildly acidic to near-neutral pH conditions.

ID	Cd(μg/L)	Fe(μg/L)	Mn(μg/L)	Ni(μg/L)	Pb(μg/L)	Zn(μg/L)	As(μg/L)	Hg(μg/L)
BH 1	0.0	7949.0	849.0	0.4	34.5	0.0	5.9	5.6
BH 2	0.0	132.4	25.8	0.0	67.9	0.0	3.0	5.1
BH 3	0.0	4187.5	235.5	20.1	33.9	0.0	3.9	5.7
BH 4	0.0	254.4	65.0	14.9	42.6	256.7	2.7	6.9
BH 6	0.0	328.1	40.1	0.0	12.4	0.0	2.9	5.3
BH 7	0.0	589.0	16.2	18.4	4.2	0.0	2.6	5.0
BH 8	0.0	213.2	9.5	0.0	23.0	11.1	3.0	5.1
BH 9	0.0	104.6	19.8	0.0	14.0	13.1	1.5	5.2
BH 10	0.0	445.0	12.8	23.2	1.8	11.2	2.3	5.0
BH 11	0.0	485.8	18.7	29.6	1.3	26.0	2.8	4.7
BH 12	0.0	1205.5	24.7	0.0	8.6	0.0	2.2	5.2
BH 13	0.0	390.2	12.5	0.0	10.3	0.0	1.7	5.0
BH 14	0.0	0.0	45.1	0.0	10.6	0.0	2.2	5.0
BH 15	0.0	0.0	50.3	7.4	2.6	67.8	1.7	4.4
BH 16	0.0	312.7	11.3	0.0	14.4	0.0	2.0	4.2
BH 17	0.0	264.1	52.0	15.9	4.8	18.4	2.4	4.4
BH 18	0.0	0.0	40.6	0.9	10.7	40.2	3.2	4.6
BH 19	0.0	436.7	7.3	25.3	3.6	32.6	3.2	4.5
BH 20	0.0	436.6	8.6	1.4	0.0	0.0	5.0	4.1
Mean	**0.01**	**933.41**	**81.31**	**8.29**	**15.85**	**25.11**	**2.85**	**5.00**

Figure 7: Concentration of the trace metals in groundwater samples collected from the study area

4.2.1 Spatial distribution of metals and contamination indices in relation to land use in the study area.

Spatial maps of heavy metal concentrations in groundwater together with the pollution indices on surface were created using ArcMap (10.8) software and were then compared to the land use in the area.

Iron (Fe)is one of the essential heavy metals for human nutrition, and it is a vital element for human life (Iron absorption). Iron is highly conserved in the body; there is no mechanism for excretion but even in such high concentrations it poses no health hazards (an introduction to iron in the body). The spatial map shows higher concentrations of Iron in the eastern part of the study area suggesting contribution from weathering of Mn-rich minerals and possibly industrial effluents (mainly from the ongoing galamsey).

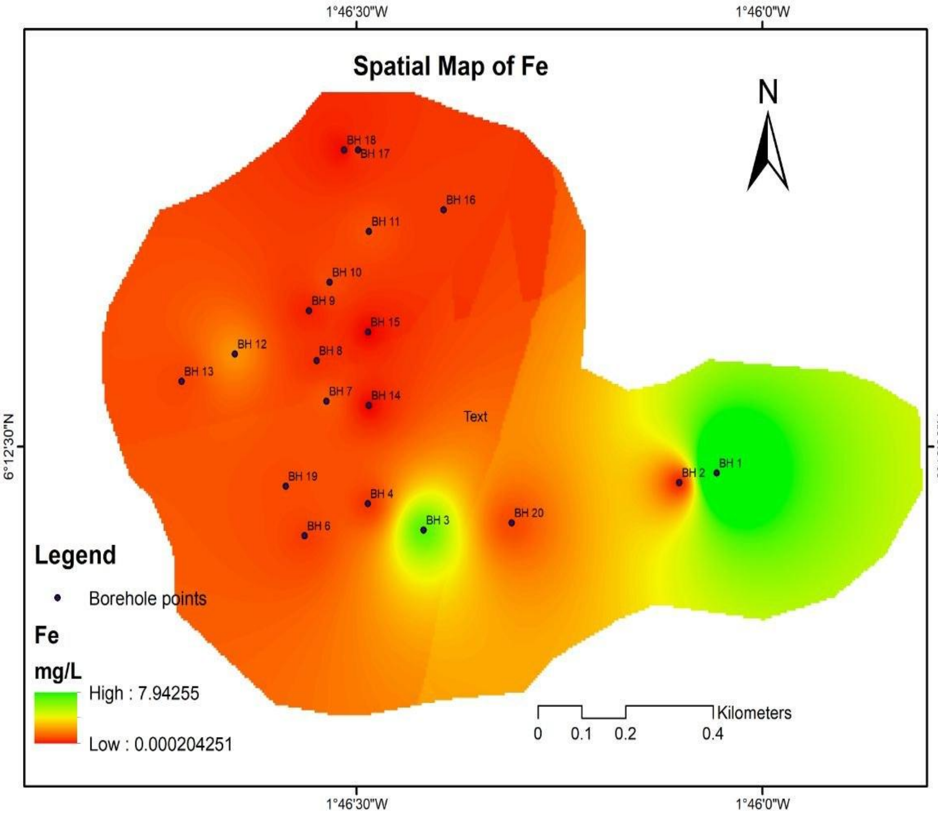


Figure 8: Spatial distribution of Iron in the study area

Lead is a naturally occurring bluish-gray metal present in small amounts in the earth's crust. Although lead occurs naturally in the environment, anthropogenic activities such as fossil fuels burning, illegal mining, and manufacturing contribute to the release of high concentrations. Lead has many different industrial, agricultural, and domestic applications. It is currently used in the production of lead-acid batteries, oxides for paint, glass, pigments and chemicals etc (heavy metal toxicity in the environment, Kana 2022). The spatial map showed higher concentrations of lead in the Eastern parts of the study area.

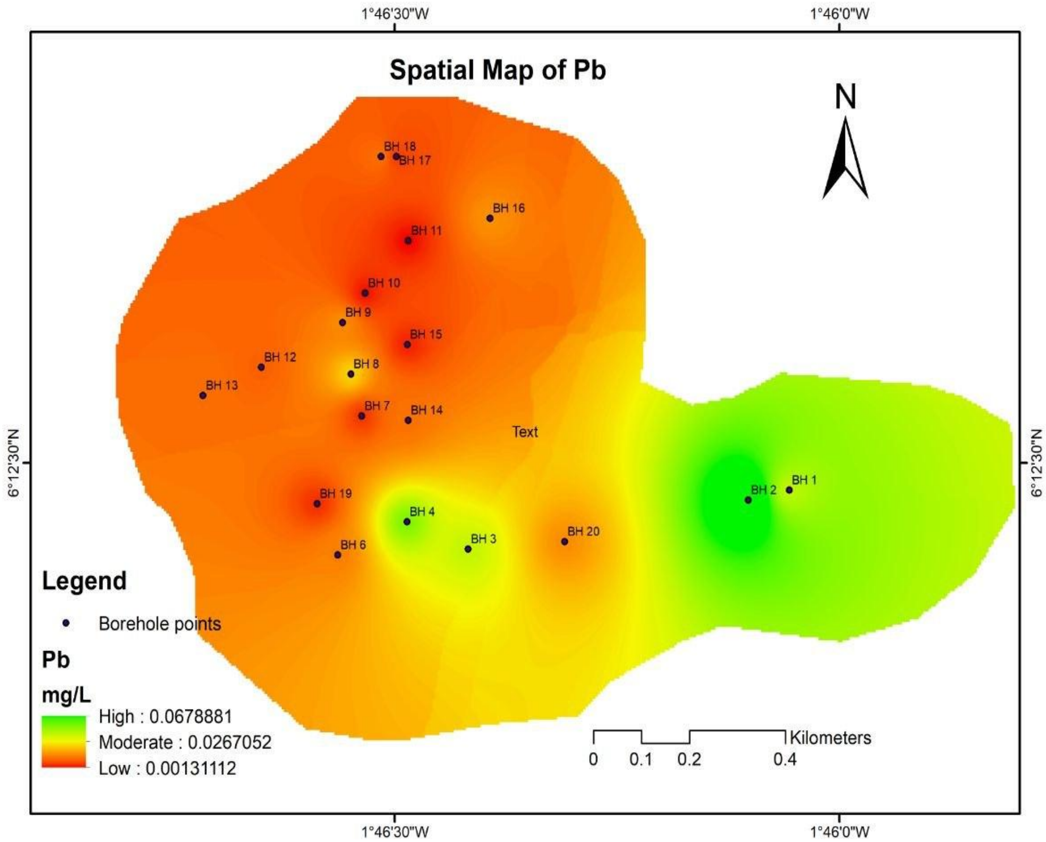
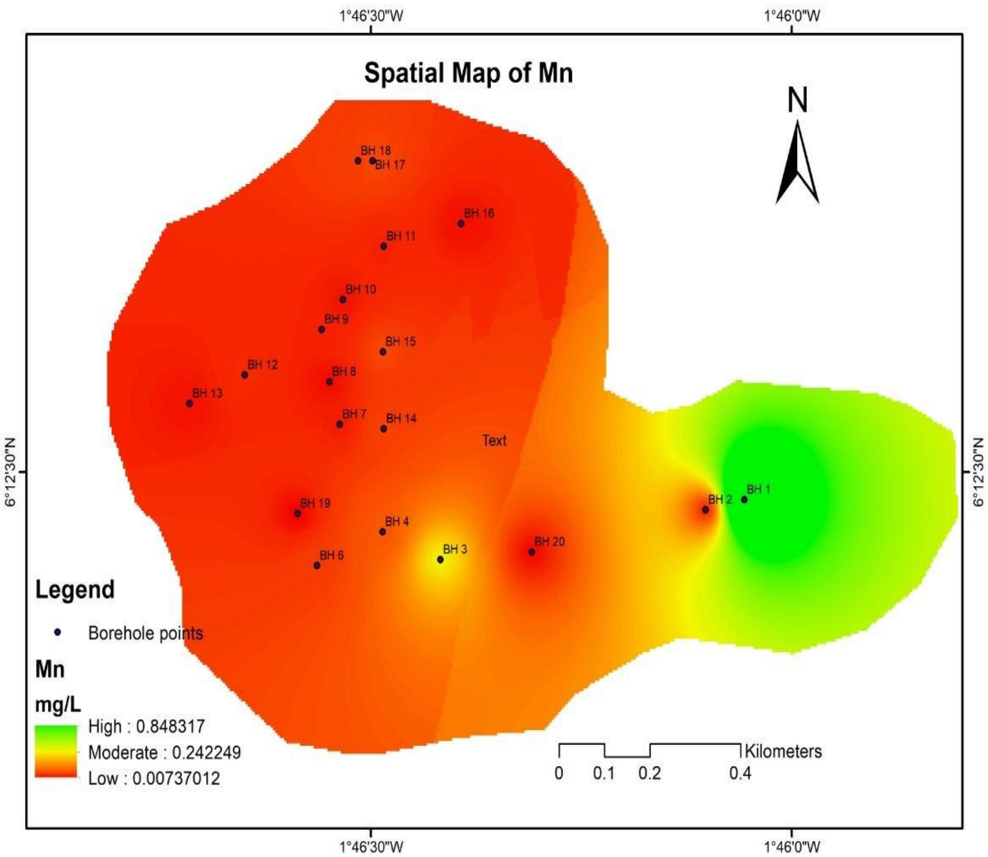


Figure 9: Spatial distribution of lead in the study area

Manganese naturally occurs in a wide variety on the surface of earth and in groundwater, in addition to soils with different rock structures and water bodies. Regrettably, a significant portion of the manganese contamination in water bodies is also caused by human activity. Manganese is mostly utilized in the manufacturing sector in the form of different manganese compounds, iron, and steel alloys, and as a component of many kinds of goods. The benefits of manganese on human health are being more widely acknowledged every day, but there are some recent studies documented that excessive manganese consumption also has significant health hazards to the people with acute neurological effects (Manganese in groundwater and its impacts on humans). Manganese showed higher concentrations in the eastern parts of the study area.



Mercury is a widespread environmental toxicant and pollutant which induces severe alterations in the body tissues and causes a wide range of adverse health effects. Humans are exposed to all forms of mercury through accidents, environmental pollution, food contamination, dental care, preventive medical practices, industrial and agricultural operations, and occupational operations (Heavy metal Toxicity and the environment).

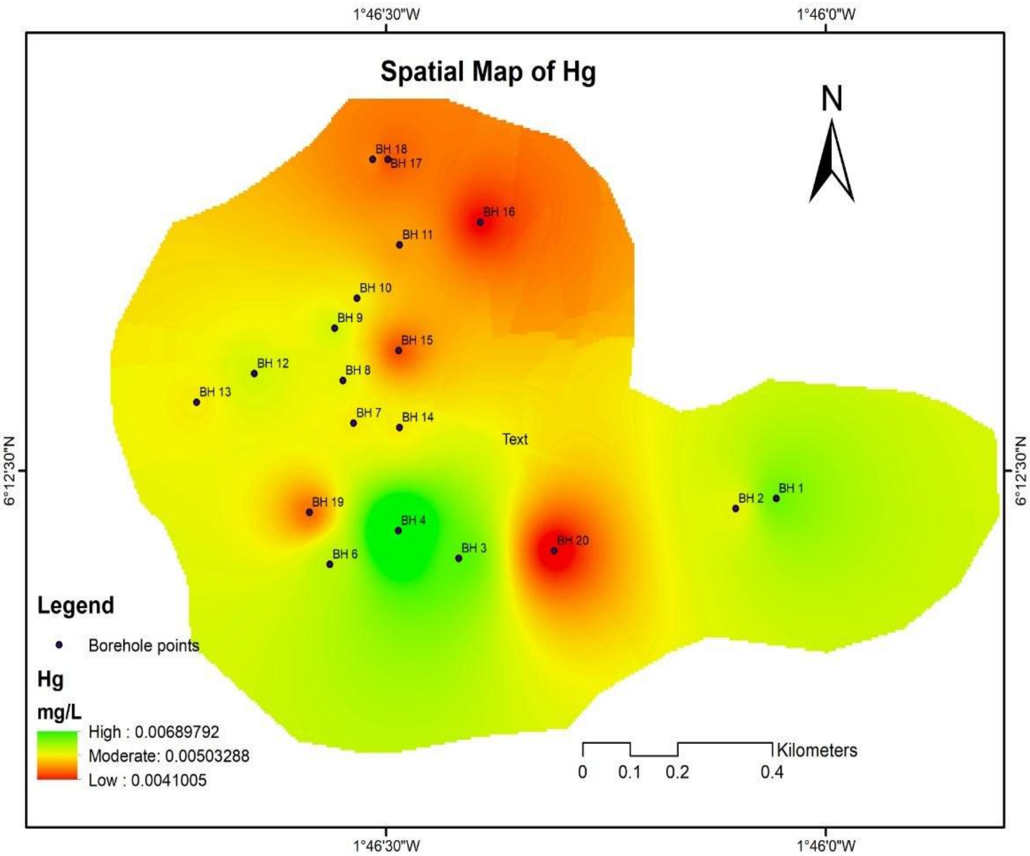


Figure 10: Spatial distribution of Mercury in the study area.

Figure 11: Ongoing galansey at the eastern side of Hia

Figure 12:Ongoing galamsey at the eastern part of Hia

Illegal mining involves the use of various harmful chemicals to extract valuable minerals, which in turn introduces heavy metals into the environment particularly groundwater. Areas with high levels of contamination should be subjected to routine monitoring, as groundwater pollution poses significant risks to public health.

WHO COMPARISON

Metal	Mean	WHO Limit (mg/L)	Status
Cd	0.00001	0.003	Within WHO permissible limit (Safe)
Fe	0.8871115	0.30	Exceeds WHO aesthetic guideline (Elevated)
Mn	0.0772405	0.08	Within WHO permissible limit (Close to limit)
Ni	0.0078795	0.07	Within WHO permissible limit (Safe)
Pb	0.016365	0.010	Exceeds WHO guideline (Unsafe)
Zn	0.0238605	3.0	Within WHO permissible limit (Safe)
As	0.00283	0.010	Within WHO permissible limit (Safe)
Hg	0.00499	0.006	**Within WHO permissible limit (Approaching limit)**

4.2.2 Heavy Metal Pollution Index.

As shown in table 3 for the classification of the heavy metal pollution index: HPI<100-low; HPI=100-Medium; HPI >100-High. Eighteen of the sample locations, representing 94.74% of the sample population had HPI less than 100, indicating low water pollution, however HPI exceeded hundred in one borehole indicating high water pollution. This high HPI value may be because of anthropogenic activities in the area which are likely to be the illegal mining (galamsey) or agricultural activities.

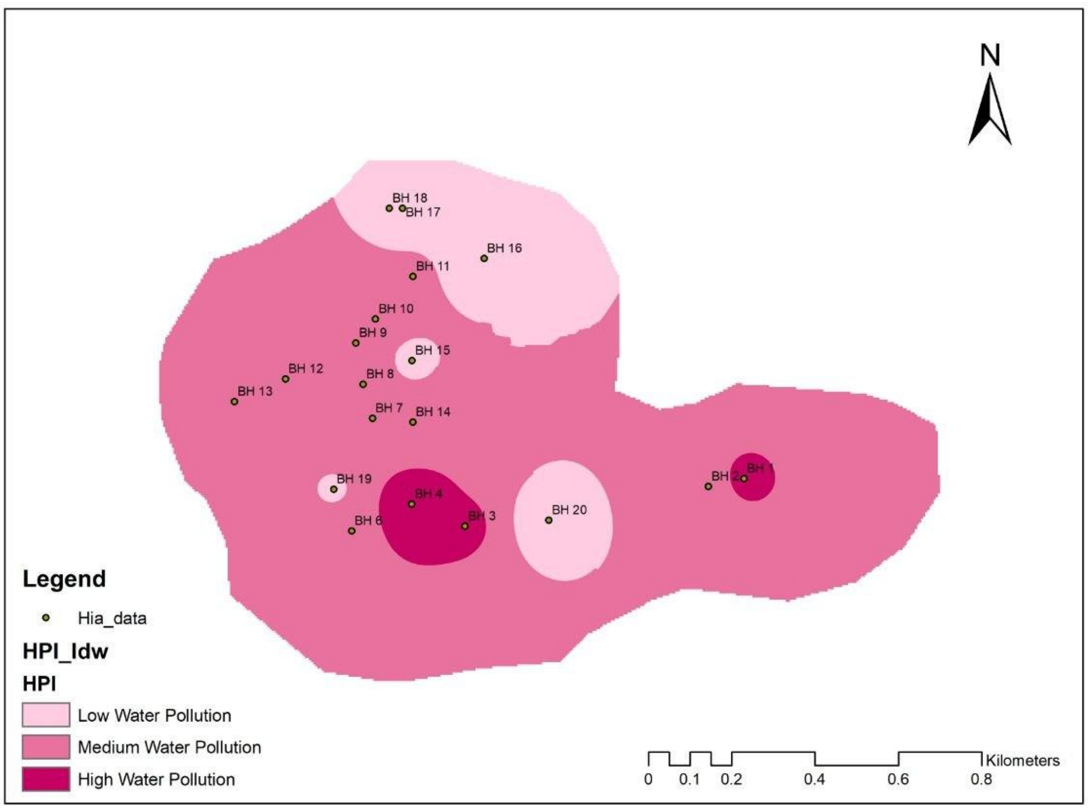


Figure 13:Spatial distribution of Heavy Metal Pollution in the study area

4.2.3 Metal Index

MI ranged from 33.60 to 4.112 with a mean of 4.40. From the table none of the samples falls within the very pure and pure category. BH15 value indicates that groundwater is slightly affected. BH19, BH18, BH17, BH16, BH14, BH13, BH11, BH10, BH9, BH7, and BH6 values fall within the moderately affected category. BH20, BH12, and BH8 values fall within the strongly affected category which may be due to anthropogenic sources possibly illegal mining (galamsey). BH4, BH3 BH2 and BH1 fall within the seriously affected category and their high concentrations may possibly be attributed to illegal mining hence these boreholes require continuous monitoring to prevent an increase in the concentrations of the already present heavy metals.

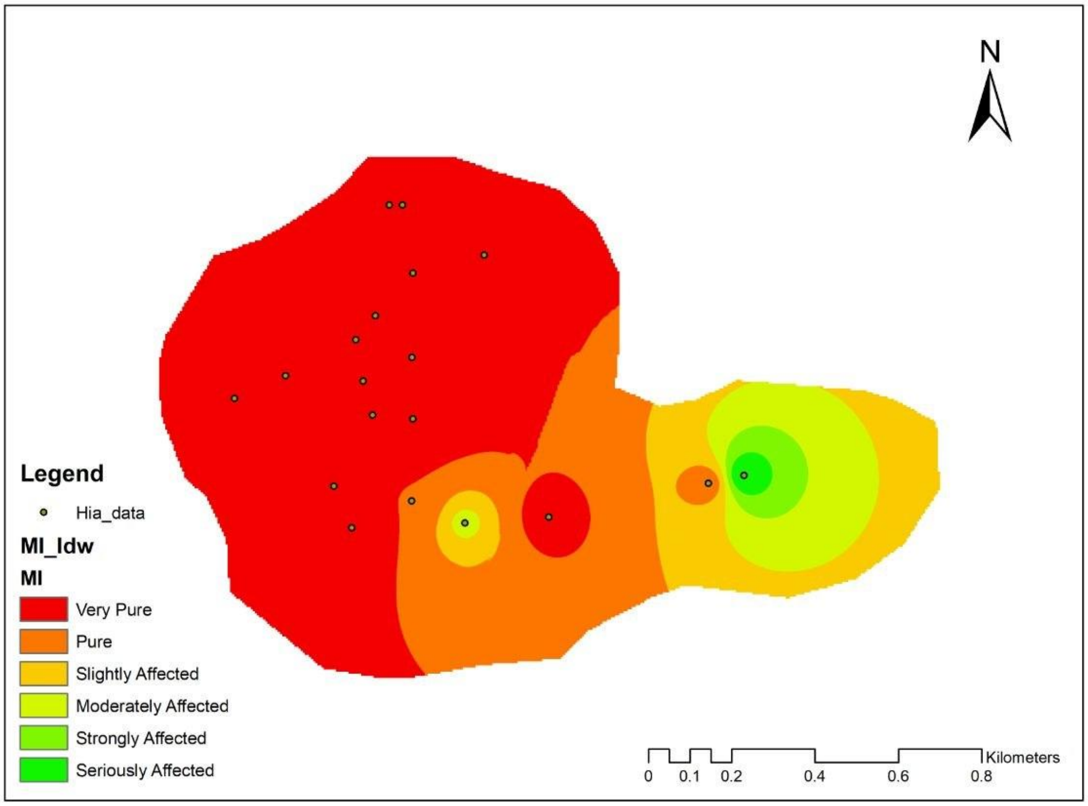


Figure 14:Spatial distribution of Metal Index in the study area

Sample ID	HPI	MI
BH 1	90.751	33.600
BH 2	118.378	8.448
BH 3	81.753	19.566
BH 4	90.433	6.991
BH 6	42.289	3.609
BH 7	30.237	3.782
BH 8	56.166	4.190
BH 9	41.819	2.821
BH 10	26.401	3.095
BH 11	25.494	3.293
BH 12	36.998	6.028
BH 13	36.624	3.367

BH 14	37.161	2.228
BH 15	23.265	1.419
BH 16	39.526	3.412
BH 17	27.939	2.699
BH 18	37.168	2.286
BH 19	28.263	3.278
BH 20	43.394	4.112
Mean	**48.108**	**6.22**

Figure 15:Groundwater quality evaluation indices for the study area

Geoaccumulation Index (Igeo)

The Wassa geochemical study reported background concentrations as range rather than single values. To obtain a representative background concentration (Bn) for the calculation of the Geoaccumulation Index, the arithmetic midpoint of each reported background range was used. This method provided a single reference value while preserving the regional geochemical characteristics of the Birimian terrain.

Table 5:Table showing background and measured concentrations of heavy metals

ID	**Cd**(μg/L)	**Fe**(μg/L)	**Mn**(μg/L)	**Ni**(μg/L)	**Pb**(μg/L)	**Zn**(μg/L)	**As**(μg/L)	**Hg**(μg/L)
BH 1	0.0	7949.0	849.0	0.4	34.5	0.0	5.9	5.6
BH 2	0.0	132.4	25.8	0.0	67.9	0.0	3.0	5.1
BH 3	0.0	4187.5	235.5	20.1	33.9	0.0	3.9	5.7
BH 4	0.0	254.4	65.0	14.9	42.6	256.7	2.7	6.9
BH 6	0.0	328.1	40.1	0.0	12.4	0.0	2.9	5.3
BH 7	0.0	589.0	16.2	18.4	4.2	0.0	2.6	5.0
BH 8	0.0	213.2	9.5	0.0	23.0	11.1	3.0	5.1
BH 9	0.0	104.6	19.8	0.0	14.0	13.1	1.5	5.2
BH 10	0.0	445.0	12.8	23.2	1.8	11.2	2.3	5.0
BH 11	0.0	485.8	18.7	29.6	1.3	26.0	2.8	4.7
BH 12	0.0	1205.5	24.7	0.0	8.6	0.0	2.2	5.2
BH 13	0.0	390.2	12.5	0.0	10.3	0.0	1.7	5.0
BH 14	0.0	0.0	45.1	0.0	10.6	0.0	2.2	5.0
BH 15	0.0	0.0	50.3	7.4	2.6	67.8	1.7	4.4
BH 16	0.0	312.7	11.3	0.0	14.4	0.0	2.0	4.2
BH 17	0.0	264.1	52.0	15.9	4.8	18.4	2.4	4.4
BH 18	0.0	0.0	40.6	0.9	10.7	40.2	3.2	4.6
BH 19	0.0	436.7	7.3	25.3	3.6	32.6	3.2	4.5
BH 20	0.0	436.6	8.6	1.4	0.0	0.0	5.0	4.1

Mean	**0.01**	**933.41**	**81.31**	**8.29**	**15.85**	**25.11**	**2.85**	**5.00**
Background conc(Bn)	0.0	28,200	195.8	10.68	7.215	37	16.985	0.49

Table 6: Igeo values for all the heavy metals

Igeo (Cd)	Igeo (Fe)	Igeo (Mn)	Igeo (Ni)	(Igeo (Pb)	Igeo (Zn)	Igeo (As)	Igeo (Hg)
-2.96347	-2.41181	1.531421	-5.32373	1.672563	-13.4383	-2.11044	2.929611
-2.96347	-8.31961	-3.5089	-11.6457	2.649378	-13.4383	-3.08619	2.794681
-2.96347	-3.3365	-0.31862	0.327321	1.647251	-13.4383	-2.70768	2.955146
-2.96347	-7.37742	-2.17583	-0.10456	1.97682	2.209524	-3.23819	3.23078
-2.96347	-7.01038	-2.87267	-11.6457	0.196306	-13.4383	-3.1351	2.850176
-2.96347	-6.16625	-4.18028	0.199832	-1.36557	-13.4383	-3.29264	2.766112
-2.96347	-7.63231	-4.95027	-11.6457	1.0876	-2.32193	-3.08619	2.794681
-2.96347	-8.65963	-3.89077	-11.6457	0.371393	-2.08292	-4.08619	2.822695
-2.96347	-6.57071	-4.52013	0.534251	-2.58796	-2.30899	-3.46952	2.766112
-2.96347	-6.44415	-3.97323	0.885723	-3.05745	-1.09398	-3.18572	2.676845
-2.96347	-5.13295	-3.57176	-11.6457	-0.33163	-13.4383	-3.53365	2.822695
-2.96347	-6.7603	-4.55434	-11.6457	-0.07139	-13.4383	-3.90562	2.766112
-2.96347	-23.0122	-2.70314	-11.6457	-0.02997	-13.4383	-3.53365	2.766112
-2.96347	-23.0122	-2.54571	-1.11428	-2.05745	0.288798	-3.90562	2.581687
-2.96347	-7.07973	-4.69995	-11.6457	0.412035	-13.4383	-3.67115	2.514573
-2.96347	-7.32343	-2.49776	-0.01085	-1.17293	-1.59278	-3.40812	2.581687
-2.96347	-23.0122	-2.85479	-4.15381	-0.01642	-0.46529	-2.99308	2.645818
-2.96347	-6.59787	-5.3303	0.659263	-1.58796	-0.76762	-2.99308	2.614109
-2.96347	-6.5982	-5.09386	-3.51638	-9.5638	-13.4383	-2.34922	2.479808

From the table, the Igeo values for Cd in all the boreholes are -2.963 that is Igeo< 0 which indicates that all the groundwater samples are unpolluted with respect to cadmium. The Fe values range from -23.01 to -2.41 approximately indicating that the groundwater samples are uncontaminated with respect to iron. The Mn values range from -5.33 to 1.53. Only one sample has Igeo of 1.53 which falls within the class of 1-2, indicating that sample 1 is moderately contaminated and all remaining samples are unpolluted. The Ni values for BH3, BH7, BH10, BH11, BH19 are 0.33, 0.20, 0.53, 0.89,0.66 respectively which falls within the Igeo class of 0 and 1 indicating that nickel is generally unpolluted, although several boreholes are classified as contaminated to moderately contaminated. Pb values range from -3.06 to 2.65 with several boreholes having concentrations exceeding 2. BH2 is found to be moderately to strongly contaminated whereas the other boreholes are moderately polluted and these high concentrations indicate both a natural source such as weathering of rock or anthropogenic source such as illegal mining or agriculture. Almost all Zn values are -13.44 except a few such as 0.29, -0.47, -0.77, 2.21 indicating that most samples are uncontaminated with only one sample having a concentration of 2.21 which is moderately to strongly contaminated. As is negative for every sample indicating that they are uncontaminated. Mercury (Hg) has values ranging from 2.48 to 3.23 which falls within the Igeo class of 2-3 and 3-4 indicating that most boreholes are moderately to strongly contaminated with a few being strongly contaminated.

4.2.4 Correlation Assessment

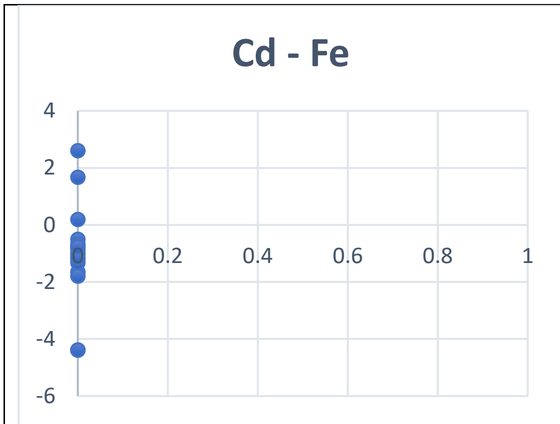
To examine the interconnections between dissolved heavy metals and their impact on groundwater quality, Pearson's bivariate correlation coefficients were computed alongside coefficient of determination (R^2) values derived from scatter plot matrices for the random twelve metal pairings shown in the figures below. The resulting R^2 values reveal an unambiguous statistical pattern: nearly all metal pairs exhibit negligible to weak explanatory power, with values ranging from a practically zero 0.0071 (As–Hg) to a relatively modest 0.2662 (Hg–Mn), while the overwhelming majority fall below the 0.20 benchmark. These findings indicate that the heavy metals in the groundwater do not share a common source, instead, they likely originate from multiple different sources that does not relate such as industrial activities, agriculture runoff, mining operation, or natural geological weathering.

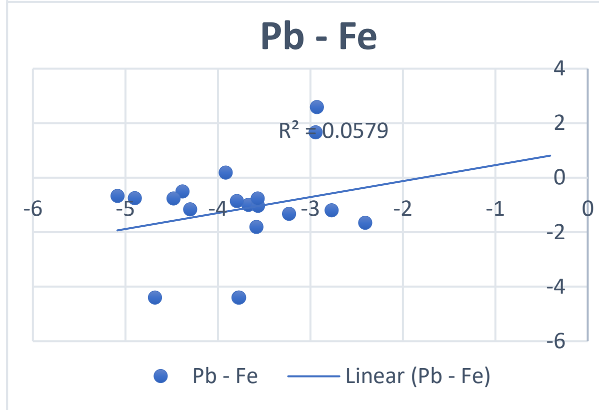
The absence of correlation also points to complex geochemical controls including pH, redox conditions, sorption, and precipitation which influence each metal's mobility and distribution differently. Cadmium (Cd) was consistently below detection limit (all zeros), so it plays no role in the pollution profile. Mercury (Hg) and Arsenic (As) show the most extreme negative values (log-normalized concentrations), suggesting they are the dominant contaminants of concern.

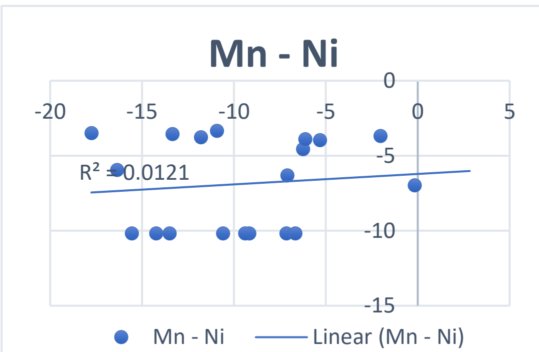
The cluster analysis groups the 19 sampling points into three distinct clusters (0, 1, and 2), revealing a clear spatial gradient in pollution intensity. Cluster 1 (samples BH 15–20) represents the least impacted background zone (lowest HPI and MI values). Cluster 2 (samples BH 2, BH 6-14) shows moderate contamination levels, while Cluster 0 (BH 1, BH 3, BH 4) reflects the most impacted background sites with concentrations of Hg, As, Mn, Pb, and Ni likely indicating a localized pollution hotspot from nearby anthropogenic activities such as artisanal gold mining or industrial discharge. Based on this clustering, a Heavy Metal Pollution

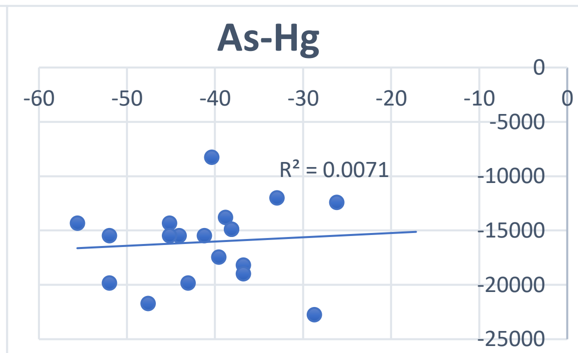
Index (HPI) and Metal Index (MI) would classify Cluster 1 as "low pollution" and Cluster 0 as "highly polluted," with Cluster 2 falling in between. These findings highlight the urgent need for targeted remediation interventions and continuous monitoring in the high-risk Cluster 0 areas to safeguard groundwater quality and public health.

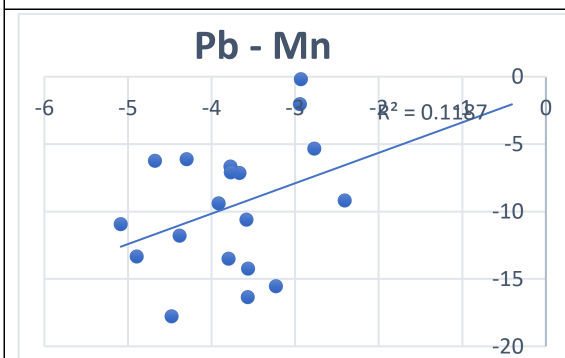
Table 7:Pearson's bivariate correlation matrix for physical parameters, heavy metal concentration and the indices of pollution.

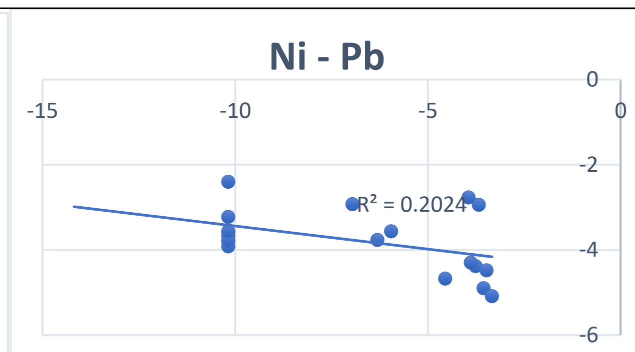


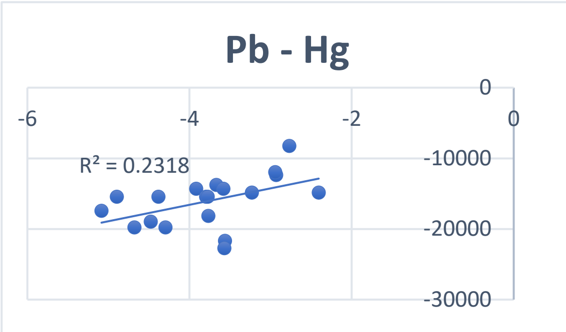


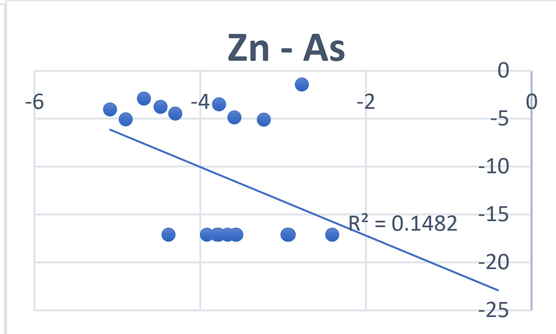


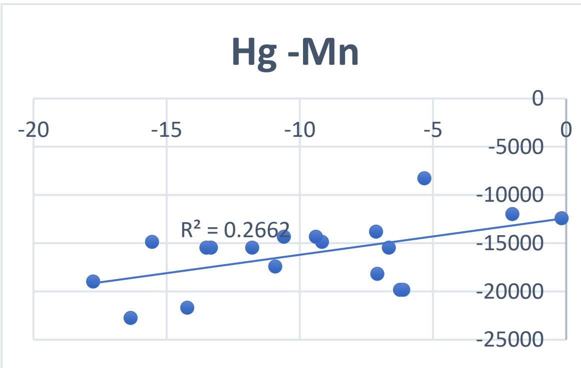


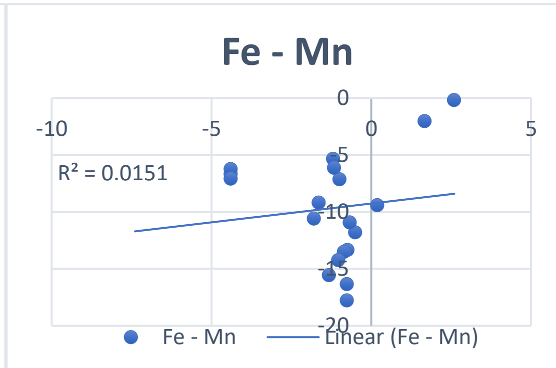


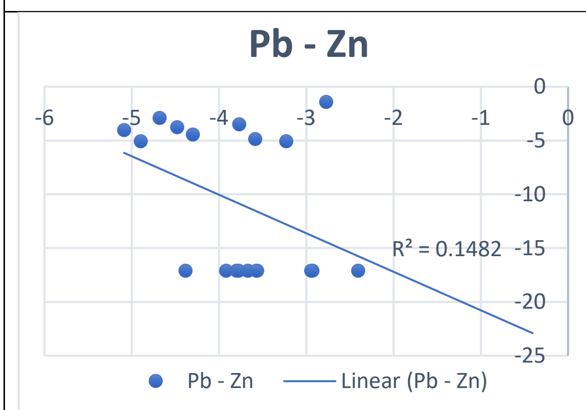


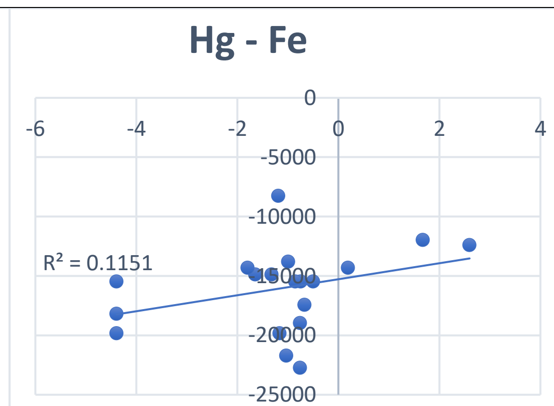

Pb - Hg
R² = 0.2318
-6
-4
-2
0
0
-10000
-20000
-30000


Zn - As
R² = 0.1482
-6
-4
-2
0
0
-5
-10
-15
-20
-25


Hg -Mn
R² = 0.2662
-20
-15
-10
-5
0
0
-5000
-10000
-15000
-20000
-25000


Fe - Mn
R² = 0.0151
-10
-5
0
5
0
-5
-10
-15
-20
Fe - Mn
Linear (Fe - Mn)


Pb - Zn
R² = 0.1482
-6
-5
-4
-3
-2
-1
0
0
-5
-10
-15
-20
-25
Pb - Zn
Linear (Pb - Zn)


Hg - Fe
R² = 0.1151
-6
-4
-2
0
2
4
0
-5000
-10000
-15000
-20000
-25000

CHAPTER 5: CONCLUSION AND RECOMMENDATION

5.1 CONCLUSION

This study evaluated groundwater quality using 19 groundwater samples collected within the study area. These water samples were analysed for physiochemical parameters as well as heavy metals (Cd, Fe, Mn, Ni, Zn, Pb, As, Hg). Heavy metal pollution index, metal index and the geoaccumulation index as well as the Cluster and the Principal Component Analysis. Descriptive statistics, pollution indices, multivariate statistical techniques and spatial interpolation were integrated to assess heavy metal contamination.

The spatial analysis revealed variations in the heavy metal concentrations in the study area with the highest concentrations seen at the eastern part of the study area.

The pollution assessment showed that only one sampling location (BH2) exceeded the recommended Heavy Metal Pollution Index (HPI) threshold of 100%, indicating localized heavy metal contamination. All the pollution indices revealed moderate to strong contamination of the heavy metals in the study area.

Principal Component Analysis indicated that the first principal component explained 43% of the overall variance, suggesting that a limited number of controlling processes account for much of the groundwater chemistry.

The results from the Correlation Assessment implies that the heavy metals found in the groundwater originated from different sources and therefore remediation cannot depend on broad spectrum treatments or simple salinity screens but must be tackled individually (metal by metal) focusing on specific point sources.

Overall, the integrated statistical and spatial analyses indicate that groundwater quality may potentially be influenced by the contamination of heavy metals introduced into the groundwater system by artisanal gold mining.

5.2 RECOMMENDATION

It is essential to establish routine groundwater quality monitoring programmes for all sampling locations to continually examine the heavy metal concentrations in these places. Prioritize locations with elevated Heavy Metal Pollution Index values for detailed investigation. Conduct seasonal groundwater monitoring to evaluate temporal variations in heavy metal concentrations. Develop groundwater protection strategies based on identified contamination hotspots. Additionally, public education on the awareness about the risks associated with heavy metals exposure as well as strict enforcements of laws concerning illegal mining 'Galamsey' should be ensured.

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